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A Palaeomagnetic Discordance Between a Lava Sequence and an Associated Interbasaltic Horizon from the Faeroe Islands*

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Summary

Demagnetization of samples covering some 250 metres of the lava series of Streymóy, Faeroe Islands, revealed the presence of a well-defined palaeomagnetic pole almost coinciding with the present pole. Results from a partly baked interbasaltic tuff horizon within the lower part of the lava series and from the weathered top section of the associated baking lava also revealed a tightly grouped distribution of directions but deviating, however, by some 18 degrees of arc from the mean of the lava sequence. Magneto-mineralogical investigations suggest that the average ' primary ' oxidation state of the weathered lava-tuff layer is considerably higher than that of the lava sequence suggesting that the magnetization acquired during weathering and baking of the former rocks has been less affected by subsequent (thermo-) chemical alterations than that of the lavas. Hence, the palaeomagnetic pole position of the tuff layer is interpreted to represent the geomagnetic field in Lower Tertiary, the time of formation of the upper lava series. The investigated lava series have been altered by low temperature oxidation which probably caused an almost complete remagnetization at about Middle Tertiary along a geomagnetic field axis coinciding with the present one.

Introduction

The Faeroe Islands which belong to the North Atlantic igneous province consist almost exclusively of subaerially extruded tholeiitic flood-basalts with an exposed stratigraphic thickness of close to 3000 m (Noe-Nygaard & Rasmussen 1968). The lava sequence has been subdivided into three parts: Upper, Middle and Lower series. Thin interbasaltic (more or less baked) tuff layers are scattered throughout the Middle and Upper basalt sequences. Radiometric age data (Tarling & Gale 1968) and palaeobotanical evidence (Laufeld 1965) indicate that the subaerial part of the Faeroe Islands was formed during Lower Eocene (50–60 Myr) times.

Previous palaeomagnetic work on lavas from the Faeroe Islands includes a reconnaissance survey for polarity zonation (Abrahamsen 1967) and a study of samples from 306 lava flows by Tarling (1970).

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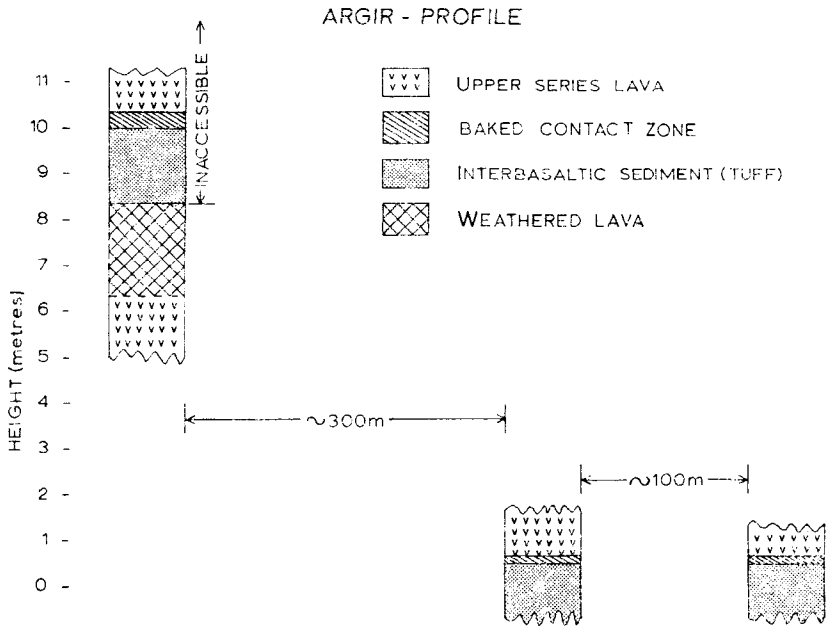


FIG. 1. Stratigraphic section of the Argir-profile.

The results of Tarling (1970) showing a scattered (less than Fisherian) distribution of directions after partial alternating field demagnetization have been ascribed to the behaviour of the geomagnetic field in Eocene times, but unsuccessful removal of secondary induced directions may be an alternative explanation. Secondary induced directions due to remagnetization processes have been established in a previous study of the Northern Irish Basalts (Løvlie, Gidskehaug & Storetvedt 1972; Wilson 1961).

Field work

A stratigraphic section of the Upper lava series covering approximately 250 m was sampled at 10 localities (45 hand samples). At Argir, some 2 km south of Tórshavn, red interbasaltic sedimentary layers are exposed within the stratigraphically lower parts of the investigated lava section. Twenty-seven drillcores were collected from a profile consisting of different rocks as outlined in Fig. 1. The same interbasaltic sedimentary layer and baked contact zone was sampled at two localities approximately 100 m apart. The overlying baking lava was sampled at two levels: just above the lower contact and in the topmost 3 m, the latter from a locality situated some 300 m from the previous site. A distinct change in colour from black in the interior of the flow to grey-brown-red in the outer 2 m or so is likely to be the result of surface alterations having taken place during and after cooling but before further lava accumulation. As will be shown below the transition from ' fresh ' to weathered lava is also accompanied by marked changes in some bulk magnetic parameters. The interbasaltic sediment layer resting upon the weathered lava was inaccessible for sampling.

Ore microscopy

Polished sections of 22 specimens from the Upper lava series and 15 specimens

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from the section across the interbasaltic layer were studied under the ore microscope at magnifications up to 1900 times. The magnetic colloid technique was used for identification purposes and to detect magnetic inhomogeneities within the magnetic grains. The results are summarized as follows:

(a) *Upper lava series.*

Ilmenite and titanomagnetite are present as grains ranging between 5 and 200 μ in size. The stratigraphically lower part of the section contains approximately equal amounts of ilmenite and titanomagnetite. The former appearing as optically homogeneous grains while the latter sometimes shows the presence of a few, thin ilmenite lamellas suggesting high-temperature oxidation state class II (Ade-Hall *et al.* 1968). Low-temperature oxidation features such as titanomagnetite (along cracks) and colour inhomogeneities (maghemite) are found within otherwise isotropic titanomagnetite grains. At higher stratigraphical levels in the lava sequence free ilmenite is the dominating ore mineral showing high temperature oxidation features such as the presence of ferri-rutile. Titanomagnetite is present in high temperature oxidation states class II-VI. The less oxidized titanomagnetite grains, class II–III, contain titanomaghemite. In this section of the lava sequence the olivine grains often appear altered due to high temperature alteration during which haematite appears to have been exsolved along grain boundaries (Haggerty & Baker 1967).

(b) *Argir profile.*

In the baking lava, near the contact to the underlying baked tuff layer, low temperature oxidation products characterize the titanomagnetite grains which belong to high temperature oxidation class I–II. The smaller grains are almost completely altered to titanomaghemite, while the larger are coated with titanohaematite with varying degrees of maghemitization within the grains. Homogeneous ilmenite is the dominating ore mineral. Higher up in this lava flow both ilmenite and titanomagnetite have suffered extensive high-temperature alteration, the latter belonging to class IV–VI. The top part of the lava, i.e. the weathered zone, have been extensively altered and the primary structures of the iron oxides have been completely erased. The major mineral phases are haematite, rutile and pseudobrookite appearing in grains with both octahedral and rhombohedral outlines.

The small grain sizes ($< 5\mu$) of the opaque minerals in the interbasaltic sediment prevent an unambiguous mineral identification. Bright, white anisotropic and magnetic grains are found scattered in the matrix of the sediment and within rock fragments. The skeletal shape of the latter grains suggests that the original titanomagnetite or ilmenite have been extensively altered to a strongly magnetic titanohematite (?) phase.

Thermomagnetic analysis

Strong field (4–6 KOe) thermomagnetic curves were obtained using a horizontal beam Curie balance, all heating and cooling being done in air. Care was taken to avoid thermal hysteresis. The results are summarized as follows:

(a) *Upper lava series*

Three different types of $J_s - T$ curves characterize this sequence: all showing irreversible changes in J_s or Curie temperature (T_c) after heating to above 600 °C. More than half of the samples give results corresponding to those of Fig. 2(a). The inflexion around 300–400 °C represents a thermally unstable magnetic mineral phase(s)

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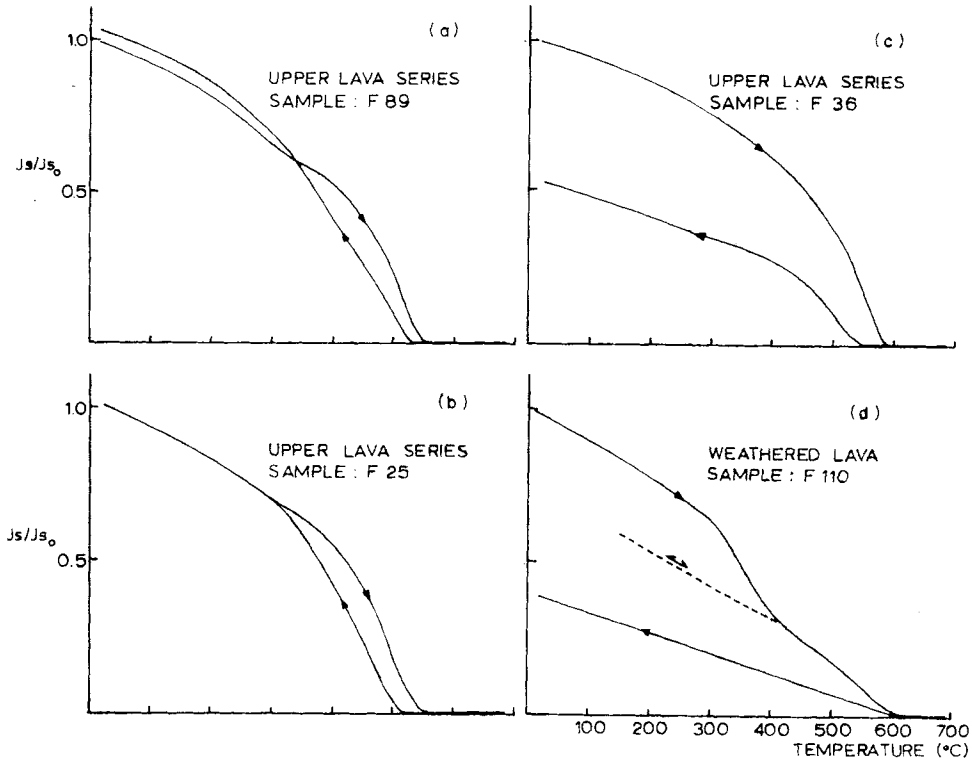


FIG. 2. Characteristic $J_s - T$ curves from the upper lava series (a, b, c) and the weathered lava section (d); broken line, cooling/heating cycle from 400° to 150 °C.

as experiments revealed an increase in J_s by cooling from this ' Curie temperature '. Isothermal heating for a few tens of minutes in this temperature range also resulted in a J_s-increase. The heating curves define Curie temperatures ranging between 510 and 570 °C, which on cooling reduces to between 480 and 560 °C associated with a 10–20 per cent increase in J_s. The process involved may be unmixing of titanomaghemite to almost pure magnetite (Readman & O'Reilly 1970). This conclusion is strengthened by the observation that whole rock and powdered specimens give almost identical $J_s - T$ curves. The thermomagnetic features shown in Fig. 2(b) probably represent the same process except that the total J_s does not change after heating.

Fourteen hand samples are presented by the $J_s - T$ curve shown in Fig. 2(c). A single initial Curie temperature between 520 and 575 °C reduced by a maximum of 30 °C after heating to above 500 °C. Almost pure magnetite or a stabilized cation deficient spinel may be associated with the initial T_c. The decrease in J_s after heating may be caused by high-temperature oxidation of magnetite to haematite: the low saturation magnetization as well as a small amount of this latter phase may be the reason why its thermomagnetic detection was unsuccessful. This may also be the reason why the titanohaematite observed in the ore microscope does not appear on the curves. The decrease of T_c after heating may be associated with a homogenization in titanomagnetite at elevated temperature (Nicholls 1955).

(b) *Argir profile*

The ' fresh ' lava specimens show the same features as illustrated in Fig. 2(c).

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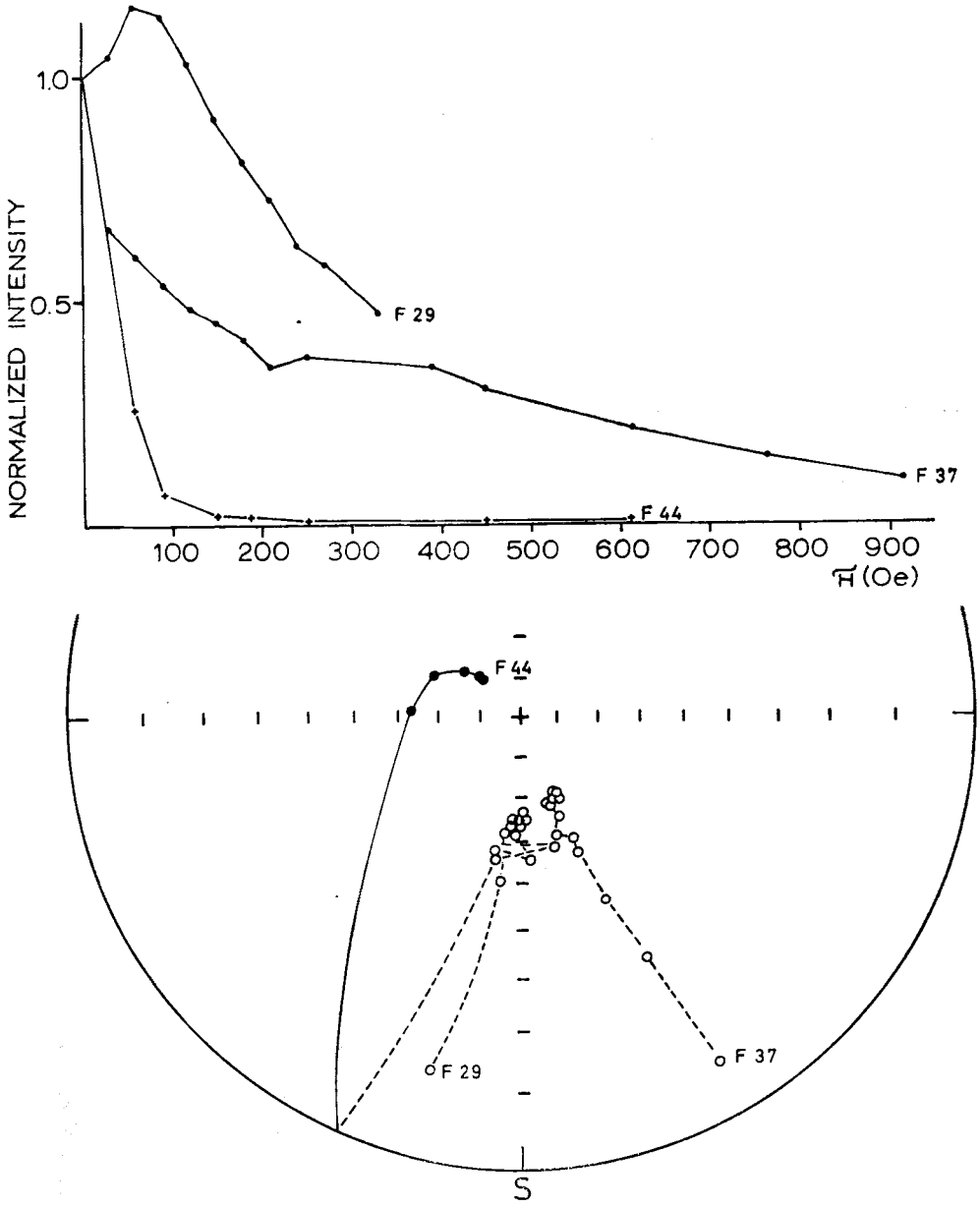


FIG. 3. Examples of variations of normalized intensity and remanence directions of upper lava samples upon progressive af-demagnetization. Filled circles, downward seeking.

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The transition from ' fresh ' to visibly weathered lava is reflected by a 30–60 per cent drop in J_s and an increase in T_c to above 600 °C. Fig. 2(d) characterizes the features observed from the weathered lava and interbasaltic specimens. The knee around 400 °C represents a transformation–temperature in that subsequent cooling after the initial heating to 400 °C produces a marked thermomagnetic irreversibility (*cf.* Fig. 2(d)). After cooling, the J_s at room temperatures has decreased by 30–50 per cent. True Curie temperatures in the 600–630 °C range which are stable after heating to 700 °C are found in samples both from the weathered lava and from the interbasaltic layer. Samples of the latter material lack the ' knee ' around 400 °C, but instead they show an almost linear decrease of J_s with temperature. X-ray diffractograms obtained from magnetic extracts prepared from the baked layer and weathered section of the lava revealed seven refraction peaks representing haematite and two refraction peaks belonging to a spinel phase similar to magnetite. The diffraction pattern did not change significantly after heating the samples in air to 700 °C. The X-ray and $J_s - T$ data together with microscope observations of a strongly magnetic, anisotropic mineral (in the baked layer as well as in the weathered lava) suggest the presence of a magnetic mineral phase in a high oxidation state which may be similar to the spinel phase formed when haematite powder is heated in water vapour to above 300 °C as reported by Johnson, Kinoshita & Merrill (1972).

Demagnetization experiments

The NRM intersities ranged between 1 and $10 \cdot 10^{-3}$ emu cm^{-3} and did not show any systematic variation with regard to rock types. Forty-six cylinders from the lava series and 15 specimens from the Argir section were subjected to progressive af-demagnetization up to maximum fields of 1200 Oe.

Some basaltic specimens exhibit large directional changes but all specimens investigated reach stable remanence directions after af-treatments to above 300 Oe.

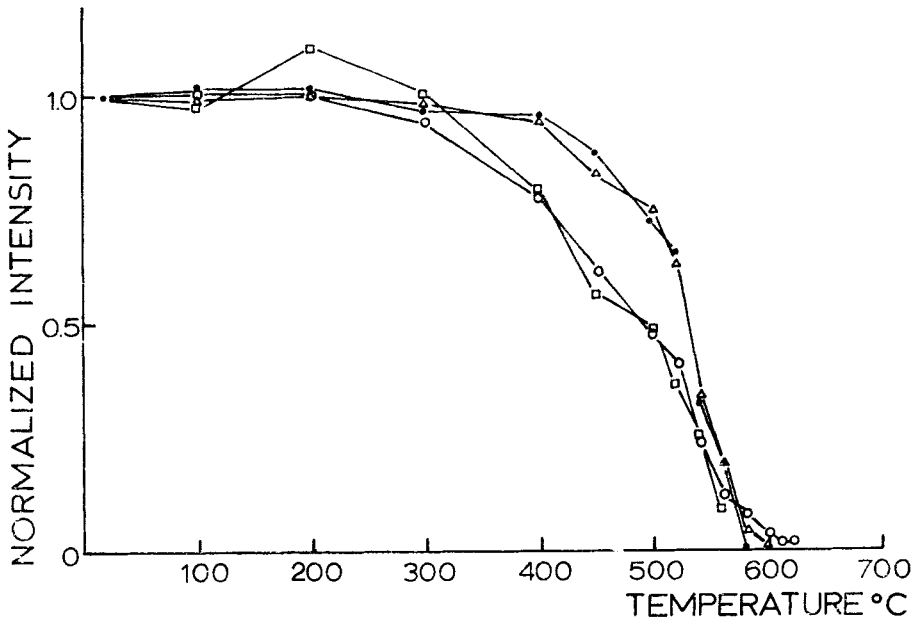


FIG. 4. Examples of normalized intensity curves of upper lava samples upon thermal demagnetization.

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Characteristic examples of directional and intensity behaviour during progressive af-demagnetization are shown in Fig. 3.

Thermal demagnetization to progressively higher temperature were performed on 21 cylinders from the lava series and 16 specimens from the Argir section. Of the first group only a few specimens showed directional changes while the latter specimens did not show any change in direction at all. The intensity of the basalt specimens remained almost constant against heat treatment to above 400° C. A major narrow blocking temperature range between 500 and 580 °C is present in most of the specimens. Fig. 4 shows examples of typical intensity-variations with progressive thermal treatment. The constancy of the intensity curves below 400 °C compared with $J_s - T$ curves (Fig. 2(a) and (b)) obtained from identical hand samples suggest that the unstable magnetic mineral associated with the ' T_c ' around 350 °C is unimportant as a carrier of remanence. This also applies to the mineral phase in the weathered lava associated with the transformation temperature around 400 °C. Stable southerly directions are found in all specimens: a fairly steep negative (upward) inclination for the lava series and somewhat lower inclination for the Argir traverse.

Evaluation of the palaeomagnetic results

Stable remanent directions of individually oriented samples from the Tórshavn lava sequence and the Argir profile are shown in Fig. 5. Mean directions, pole position and other relevant statistical parameters (Fisher 1952) are presented in Table 1. At the α_{95}-level the calculated mean direction of the upper lava series and that of the weathered lava-sediment section at Argir define two significantly different palaeomagnetic vectors deviating by some 18 degrees of arc. The less that 5° radii of the α_{95}-circles of confidence and the relatively high precision ($\kappa > 50$) of the distributions may be interpreted in terms of spot readings of the geomagnetic field. This interpretation apparently applies to the baked layer in which a TRM or a TCRM may have been induced during the relatively short time of baking by the overlying (8 m thick) lava flow. The top portion of the baking lava probably reflects a period of extensive surface alteration of the flow prior to the explosive onset of the following lava (*cf.* Fig. 1). Macroscopically the weathered lava section appears hard and compact and it is likely that the overlying tuff layer has acted as an effective thermal insulator against heat conduction associated with cooling of the overlying flow, i.e. the remanent magnetization acquired during surface weathering of the lava flow (CRM) was probably not affected by significant increase in temperature when the subsequent lava was laid down. It is thus reasonable to assume that the weathered lava acquired a chemical remanence during a somewhat more extensive time interval. This probably implies that the palaeomagnetic direction residing in the weathered lava zone represents a mean direction of the palaeomagnetic field (Storetvedt 1971). There is, however, no significant directional difference between the baked contact/tuff layer (' spot-reading ') and the overlying weathered lava (' average direction '), implying small directional variations of the ambient field during a time span comparable to the acquisition time of the supposed CRM in the weathered zone.

The mean direction of the baking lava is not significantly different from the mean of the top section of the weathered lava flow and the underlying sedimentary rocks, although the inclination is slightly steeper. Of all the lavas investigated, the lower part of the baking lava appears to have suffered the most extensive alterations due to low temperature processes taking place when the lava flows came to rest on top of the relatively porous (air/water filled) tuff layer. Thus, the ' primary ' higher than average oxidation state of this lava flow may be the reason why the late deuteric magnetization has resisted further alterations. The mean direction of the investigated lava sequence represents a lava pile some 250 m in thickness; a few sites near the

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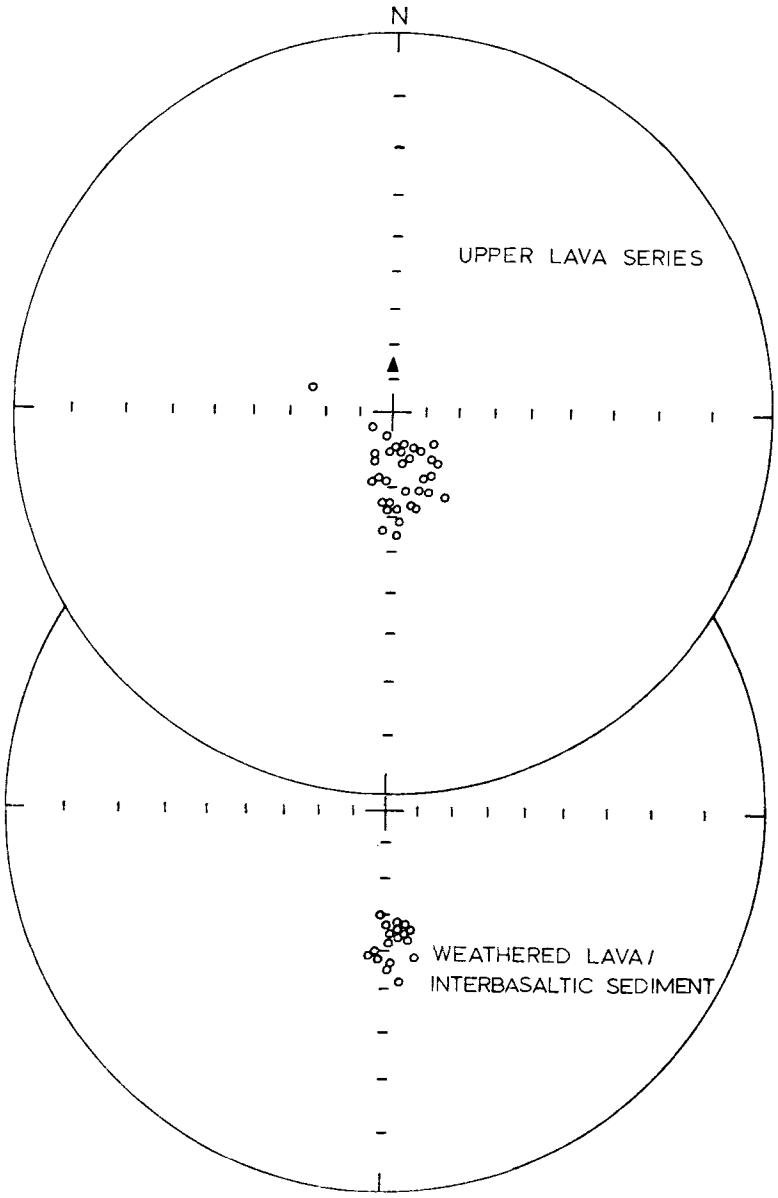


FIG. 5. Stable magnetic directions of separate samples from the upper lava series and the Argir profile (weathered lava/interbasaltic sediment). All directions pointing upwards. Filled triangle, present field.

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Table I

Palaeomagnetic directions and pole positions

	N	D	I	α_{95}	κ	λ	ϕ
Upper lava series, Tórshavn region	34	171·9	−72·2	3·5	53·4	218° E	84° N
Weathered baking lava/sediment, Argir	18	175·1	−53·9	2·2	257·6	182° E	62° N
Baking lava, Argir	9	169·0	−58·8	4·1	159·9	—	—
Interbasaltic rocks, N. Ireland (Wilson 1961)	8	178·0	−41·0	8·0*	—	175° E	60° N

N, number of separate samples;
D, mean declination;
I, mean inclination;
α_{95}, radii of 95 per cent circle of confidence;
κ, precision parameter;
λ, longitude of axial palaeomagnetic pole position;
ϕ, latitude of axial palaeomagnetic pole position;
*, standard deviation.

base of the sequence cover the same stratigraphic level as the Argir section. The variation of stability against thermal and af-demagnetization may reflect both the range of deuteric oxidation states (Strangway, Larson & Goldstein 1968; Larson *et al.* 1969), as well as the degree of low temperature alterations of the titanomagnetite grains (Johnson & Merrill 1973). All samples reveal a T_c in the 500–580 °C range; apparently in agreement with the range of deuteric oxidation states of the titanomagnetite grains. It has been demonstrated (Ade-Hall, Palmer & Hubbard 1971), however, that regional hydrothermal activity may alter the magnetic properties of titanomagnetite in low deuteric oxidation states to such an extent that thermally stable magnetic phases with a single T_c above 500 °C may appear. Stable Curie temperatures in the 520–580 °C range may thus reflect the presence of either almost pure magnetite formed as a result of high/(low?) temperature oxidation or a stabilized titanomaghemite (cation deficient spinel formed at low temperatures). It has been suggested that low temperature oxidation processes may play an important role in the acquisitions of a palaeomagnetic record in lavas (Storetvedt 1973). In the present case rocks containing titanomagnetite grains belonging to different high/low temperature oxidation states reveal a fairly tight distribution of stable palaeomagnetic directions, deviating significantly from directions residing in the investigated weathered lava/sediment profile in which the magnetic minerals occur in a much higher oxidation state. It seems, therefore, reasonable to suggest that the lava sequence concerned has been completely remagnetized at a time when the geomagnetic axis almost coincided with that of the present time. The higher ' primary ' oxidation states found in the Argir section made these rocks rather resistent against secondary magnetic alterations revealing a primary palaeomagnetic direction representative of the Lower Tertiary field.

This conclusion is strengthened by the fact that the baked layer section at Argir gives a pole position almost coinciding with the proposed best estimate of the Lower Tertiary pole relative to Northern Ireland (Løvlie, Gidskehaug & Storetvedt 1972) derived from bauxite and baked laterites. This latter argument also seems to rule out the possibility that the directions represent spot-readings, since otherwise not the same geomagnetic field vector should have been memorized in highly oxidized interbasaltic layers within lava sequences extruded probably at different times in different geographic areas.

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